

Def.

Let $f: [a, b] \rightarrow \mathbb{C}$ be a function. Given a partition $P = \{x_0, \dots, x_n\}$, define

$$V(f, P) \stackrel{\text{def}}{=} \sum_{i=1}^n |f(x_i) - f(x_{i-1})|$$

which is called the variation of f with respect to a partition P . And, the value

$$TV(f) \stackrel{\text{def}}{=} \sup \{V(f, P) \mid P \text{ is partition on } [a, b]\}$$

is called the total variation of f

If the total variation is finite, we say f is of bounded variation.

Prop.

If $f: [a, b] \rightarrow \mathbb{C}$ is continuously differentiable, then f is of bounded variation with

$$TV(f) = \int_a^b |f'(t)| dt$$

Proof. Given a partition $P = \{x_0, \dots, x_n\}$ on $[a, b]$, by the fundamental theorem of calculus,

$$V(f, P) = \sum_{i=1}^n |f(x_i) - f(x_{i-1})| = \sum_{i=1}^n \left| \int_{x_{i-1}}^{x_i} f'(t) dt \right| \leq \sum_{i=1}^n \int_{x_{i-1}}^{x_i} |f'(t)| dt = \int_a^b |f'(t)| dt$$

Since P was arbitrary, $TV(f) \leq \int_a^b |f'(t)| dt$, so the total variation has finite value.

To show that the reverse inequality, we use the fact that:

f' is continuous uniformly, because it is continuous on the compact set $[a, b]$.

Given $\varepsilon > 0$, take $\delta_1 > 0$ such that $|x - t| < \delta_1 \Rightarrow |f'(x) - f'(t)| < \varepsilon$.

And, take $\delta_2 > 0$ such that if a partition P on $[a, b]$ satisfies $\|P\| < \delta_2$, then

$$\left| \int_a^b |f'(t)| dt - \sum_{i=1}^n |f'(z_i)| \cdot |x_{i+1} - x_i| \right| < \varepsilon$$

where $P = \{x_0, \dots, x_n\}$ and $z_i \in [x_i, x_{i+1}]$ is arbitrary. The existence of δ_2 is given by the fact that $|f'|$ is Riemann-integrable.

Prop. Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded variation.

Then, for any $x \in [a, b)$, the right-sided limit $f(x+)$ exists.

Similarly, for any $x \in (a, b]$, the left-sided limit $f(x-)$ exists.

Proof. Let $x \in [a, b)$ be given. Since f is of bounded variation on $[a, b]$, f is also of bounded variation on $[x, b]$. Let $M = TV_x^b(f)$, the total variation.

Suppose that $f(x+)$ does not exist. That is, for any value $L \in \mathbb{R}$, there exists $\epsilon > 0$ such that for all $\delta > 0$, there is $x < t < x + \delta$ such that

$$|f(t) - L| \geq \epsilon.$$

Let $P = \{x_0, x_1, \dots, x_n\}$ be a partition on $[x, b]$ where $x_0 = x$, $x_n = b$ and $x_0 < \dots < x_n$.